

Cofinality and Inaccessible Cardinals

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Cofinality

Recall two basic facts about cardinalities:

- Finite unions of finite sets are finite.
- Countable unions of countable sets are countable (using AC).

This means it is “hard” to reach the cardinals $\aleph_0$ and $\aleph_1$ “from below”: you cannot obtain a set of size $\aleph_0$ by putting together a few small pieces, and similarly for $\aleph_1$. The notion of **cofinality** quantifies “how hard” it is to reach an ordinal from below.

Throughout this section, α denotes a limit ordinal (following Kunen, Def. I.13.10, we define cofinality only for limit ordinals).

Definition 0.1.

A set $X \subseteq \alpha$ is **cofinal** in α if

$$\forall \gamma < \alpha \exists \xi \in X \quad \gamma < \xi.$$

Equivalently,

$$\alpha = \bigcup_{\xi \in X} \xi = \sup X.$$

Definition 0.2.

The **cofinality** of α is

$$\text{cf}(\alpha) = \text{least ordinal } \beta \text{ such that there exists } f : \beta \rightarrow \alpha \text{ with } \text{ran}(f) \text{ cofinal in } \alpha.$$

We call such an f a **cofinal map**.

We can always assume that a cofinal map $f : \text{cf}(\alpha) \rightarrow \alpha$ is **strictly increasing**: if necessary, we can “thin it out”.

Lemma 0.1.

There exists a strictly increasing cofinal map $g : \text{cf}(\alpha) \rightarrow \alpha$.

Proof. Let $f : \text{cf}(\alpha) \rightarrow \alpha$ be cofinal. Define g by recursion:

$$g(\eta) = \max \{f(\eta), \sup\{g(\zeta) + 1 : \zeta < \eta\}\}.$$

We have to check that $g(\eta) < \alpha$. If $\sup\{g(\zeta) + 1 : \zeta < \eta\} = \alpha$, then $g \upharpoonright_\eta$ would be a cofinal map from $\eta < \text{cf}(\alpha)$ into α (here we use that α is a limit), contradicting the minimality of $\text{cf}(\alpha)$. By construction, g is strictly increasing, and $g(\eta) \geq f(\eta)$ implies that g is cofinal. $\square$

Example 0.1.

- $\text{cf}(\omega) = \omega$
- $\text{cf}(\omega + \omega) = \omega$, via $n \mapsto \omega + n$
- $\text{cf}(\omega \cdot \omega) = \omega$, via $n \mapsto \omega \cdot n$

Proposition 0.1.

For any limit ordinal α ,

1. $\text{cf}(\alpha) \leq \alpha$,
2. $\text{cf}(\alpha)$ is a cardinal,
3. $\text{cf}(\text{cf}(\alpha)) = \text{cf}(\alpha)$.

Proof.

- (1) The identity map $\alpha \rightarrow \alpha$ is cofinal.
- (2) Let $\beta = \text{cf}(\alpha)$ and suppose $|\beta| < \beta$. Let $h : |\beta| \rightarrow \beta$ be a bijection and $f : \beta \rightarrow \alpha$ cofinal. Then $f \circ h : |\beta| \rightarrow \alpha$ has the same range as f and hence is cofinal, contradicting the minimality of β .
- (3) By (1), $\text{cf}(\text{cf}(\alpha)) \leq \text{cf}(\alpha)$. For the other inequality, let $f : \text{cf}(\alpha) \rightarrow \alpha$ be strictly increasing and cofinal (Lemma 0.1), and let $g : \delta \rightarrow \text{cf}(\alpha)$ be cofinal. Then $f \circ g : \delta \rightarrow \alpha$ is cofinal: given $\gamma < \alpha$, pick η with $f(\eta) > \gamma$, and $\zeta < \delta$ with $g(\zeta) > \eta$. Then $f(g(\zeta)) > f(\eta) > \gamma$. Hence $\delta \geq \text{cf}(\alpha)$.

$\square$

Regular and singular cardinals

Recall the enumeration of all infinite cardinals by means of the $\aleph$ -sequence:

$$\aleph_0 = \omega, \quad \aleph_{\alpha+1} = \aleph_\alpha^+, \quad \aleph_\lambda = \sup\{\aleph_\xi : \xi < \lambda\} \text{ for limit } \lambda.$$

Here κ^+ is the least cardinal $> \kappa$. Some cardinals are limits of short sequences of smaller cardinals – for example,

$$\aleph_\omega = \lim_n \aleph_n$$

is uncountable, but a limit of a countable sequence of smaller cardinals.

Definition 0.3.

Let κ be an infinite cardinal. κ is

- **regular** if $\text{cf}(\kappa) = \kappa$,
- **singular** if $\text{cf}(\kappa) < \kappa$.

In other words, a regular cardinal κ cannot be reached by less than κ -many steps:

$$\text{reg}(\kappa) \iff \forall \alpha < \kappa \forall f (f : \alpha \rightarrow \kappa \rightarrow \sup_{\xi < \alpha} f(\xi) < \kappa).$$

Example 0.2.

- $\aleph_0$ is regular.
- $\aleph_1$ is regular.
- $\aleph_\omega$ is singular, with $\text{cf}(\aleph_\omega) = \omega$, as witnessed by the cofinal sequence $\aleph_0, \aleph_1, \aleph_2, \dots \rightarrow \aleph_\omega$.

The regularity of $\aleph_1$ is an instance of a general fact.

Proposition 0.2.

(AC) *Every infinite successor cardinal is regular.*

Proof. Suppose $\delta = \text{cf}(\aleph_{\alpha+1}) < \aleph_{\alpha+1}$. Then there exists a cofinal $f : \delta \rightarrow \aleph_{\alpha+1}$, so

$$\aleph_{\alpha+1} = \bigcup_{\gamma < \delta} f(\gamma).$$

Now $|\delta| < \aleph_{\alpha+1}$, and $|f(\gamma)| < \aleph_{\alpha+1}$ for all $\gamma < \delta$. Hence $|\delta| \leq \aleph_\alpha$ and $|f(\gamma)| \leq \aleph_\alpha$ for all $\gamma < \delta$. This implies

$$|\aleph_{\alpha+1}| = \left| \bigcup_{\gamma < \delta} f(\gamma) \right| \leq \aleph_\alpha \cdot \aleph_\alpha = \aleph_\alpha,$$

a consequence of Hessenberg's Theorem (see the notes on Cardinals). This is a contradiction. $\square$

On the other hand,

$$\aleph_\omega, \aleph_{\omega+\omega}, \aleph_{\aleph_\omega}, \aleph_{\aleph_{\aleph_\omega}}, \dots$$

are all singular. This follows from the following result, which reduces the cofinality of $\aleph_\lambda$ to that of its index.

Proposition 0.3.

If λ is a limit ordinal, then $\text{cf}(\aleph_\lambda) = \text{cf}(\lambda)$.

Proof. ($\leq$) Let $f : \gamma \rightarrow \lambda$ be cofinal. Then

$$\hat{f} : \gamma \rightarrow \aleph_\lambda, \quad \beta \mapsto \aleph_{f(\beta)}$$

is cofinal in $\aleph_\lambda$: If $\eta < \aleph_\lambda$, then $\eta < \aleph_\xi$ for some $\xi < \lambda$. Pick β with $f(\beta) > \xi$. Then $\aleph_{f(\beta)} > \aleph_\xi > \eta$.

($\geq$) Let $f : \gamma \rightarrow \aleph_\lambda$ be cofinal. We turn f into a cofinal map $g : \gamma \rightarrow \lambda$ by letting

$$g(\beta) = \text{the } \alpha \text{ such that } |f(\beta)| = \aleph_\alpha$$

(and $g(\beta) = 0$ if $f(\beta)$ is finite). Since $f(\beta) < \aleph_\lambda$, $g(\beta) < \lambda$. If $\xi < \lambda$, pick β with $f(\beta) > \aleph_\xi$. Then $|f(\beta)| \geq \aleph_\xi$, so $g(\beta) \geq \xi$. Hence g is cofinal in λ . $\square$

For example, $\text{cf}(\aleph_{\omega+\omega}) = \text{cf}(\omega + \omega) = \omega$.

Cofinality of the continuum

Cofinality also puts a restriction on the possible values of 2^κ . To prove this, we need a fundamental inequality between cardinal sums and products. For a family of cardinals $(\kappa_i)_{i \in I}$, let

$$\sum_{i \in I} \kappa_i = \left| \bigcup_{i \in I} \{i\} \times \kappa_i \right|, \quad \prod_{i \in I} \kappa_i = |\{g : g \text{ function, } \text{dom}(g) = I, \forall i \ g(i) \in \kappa_i\}|.$$

Theorem 0.1 (König's Theorem).

(AC) Let $(\kappa_i)_{i \in I}$ and $(\lambda_i)_{i \in I}$ be families of cardinals such that $\kappa_i < \lambda_i$ for all $i \in I$. Then

$$\sum_{i \in I} \kappa_i < \prod_{i \in I} \lambda_i.$$

Proof. The map $(i, \xi) \mapsto g_{i, \xi}$, where $g_{i, \xi}(i) = \xi + 1$ and $g_{i, \xi}(j) = 0$ for $j \neq i$, is an injection, so $\sum \kappa_i \leq \prod \lambda_i$.

Suppose $f : \bigcup_i \{i\} \times \kappa_i \rightarrow \prod_i \lambda_i$. We show that f is not onto. For each $i \in I$, the set

$$\{f(i, \xi)(i) : \xi < \kappa_i\} \subseteq \lambda_i$$

has cardinality $\leq \kappa_i < \lambda_i$, so we can let $y(i)$ be the least element of λ_i not in this set. Then $y \in \prod_i \lambda_i$, but $y \neq f(i, \xi)$ for all i, ξ , since y and $f(i, \xi)$ differ at coordinate i .

(The Axiom of Choice is needed to ensure that the sets on both sides are well-orderable, so that their cardinalities are defined; for families of arbitrary sets, one uses AC to choose the witnesses $y(i)$.) $\square$

Proposition 0.4.

If $\kappa \geq \aleph_0$, then $\text{cf}(2^\kappa) > \kappa$.

Proof. Suppose $f : \delta \rightarrow 2^\kappa$ is cofinal with $\delta \leq \kappa$, so that

$$2^\kappa = \bigcup_{\gamma < \delta} f(\gamma).$$

Then $|f(\gamma)| < 2^\kappa$ for all $\gamma < \delta$, and by König's Theorem,

$$\begin{aligned} 2^\kappa &= \left| \bigcup_{\gamma < \delta} f(\gamma) \right| \leq \sum_{\gamma < \delta} |f(\gamma)| \\ &< \prod_{\gamma < \delta} 2^\kappa = (2^\kappa)^\delta = 2^{\kappa \cdot \delta} = 2^\kappa, \end{aligned}$$

where the last equality uses $\delta \leq \kappa$. This is a contradiction. $\square$

Corollary 0.1.

$2^{\aleph_0} \neq \aleph_\omega$.

Proof. $\text{cf}(2^{\aleph_0}) > \aleph_0 = \text{cf}(\aleph_\omega)$. $\square$

Inaccessible cardinals

We have seen that successor cardinals are regular, and that $\aleph_\lambda$ for “small” limits λ are singular. This suggests the question:

 Question

Are there regular limit cardinals, i.e. regular cardinals of the form $\aleph_\lambda$ with λ limit?

If $\aleph_\lambda$ is a regular limit cardinal, then by Proposition 0.3 and Proposition 0.1,

$$\aleph_\lambda = \text{cf}(\aleph_\lambda) = \text{cf}(\lambda) \leq \lambda,$$

and since always $\lambda \leq \aleph_\lambda$, we must have $\aleph_\lambda = \lambda$. So a regular limit cardinal must be very large: it is a fixed point of the $\aleph$ -function.

This is captured by the notion of **inaccessibility**.

Definition 0.4 (Hausdorff 1908, Tarski, Zermelo 1930).

An uncountable cardinal $\kappa > \omega$ is

- **weakly inaccessible** if

$$\begin{aligned} \text{reg}(\kappa) \wedge \exists \lambda (\text{lim}(\lambda) \wedge \kappa = \aleph_\lambda) \\ (\Leftrightarrow \text{reg}(\kappa) \wedge \forall \alpha < \kappa \ \alpha^+ < \kappa) \end{aligned}$$

- **(strongly) inaccessible** if

$$\text{reg}(\kappa) \wedge \forall \alpha < \kappa \ 2^\alpha < \kappa$$

Every strongly inaccessible cardinal is weakly inaccessible, since $\alpha^+ \leq 2^\alpha$. Under the **Generalized Continuum Hypothesis**,

$$(\text{GCH}) \quad \forall \alpha \ 2^{\aleph_\alpha} = \aleph_{\alpha^+}$$

weakly and strongly inaccessible cardinals coincide.

If κ is (weakly) inaccessible, then $\kappa = \aleph_\kappa$, as shown above.

Inaccessibles and the von Neumann hierarchy

The cardinality of V_α grows rather fast relative to α . For example,

$$|V_{\omega+\alpha}| = \beth_\alpha$$

where the **beth function** $\beth_\alpha$ is defined as

$$\begin{aligned} \beth_0 &= \aleph_0 \\ \beth_{\alpha+1} &= 2^{\beth_\alpha} \\ \beth_\lambda &= \sup\{\beth_\alpha : \alpha < \lambda\} \quad \lambda \text{ limit} \end{aligned}$$

The existence of **inaccessible cardinals** ensures that the von Neumann hierarchy is “long enough” for α to eventually “catch up” with the cardinality of V_α .

Proposition 0.5.

If κ is strongly inaccessible, then $|V_\kappa| = \kappa$.

Proof. Since $\kappa \subseteq V_\kappa$, $|V_\kappa| \geq \kappa$. For the other inequality, it suffices to show that $|V_\alpha| < \kappa$ for all $\alpha < \kappa$, since then $|V_\kappa| = |\bigcup_{\alpha < \kappa} V_\alpha| \leq \kappa \cdot \kappa = \kappa$.

We show $|V_\alpha| < \kappa$ by induction on $\alpha < \kappa$: if $|V_\alpha| < \kappa$, then $|V_{\alpha+1}| = 2^{|V_\alpha|} < \kappa$ since κ is strongly inaccessible. If $\lambda < \kappa$ is a limit, then V_λ is a union of $|\lambda| < \kappa$ many sets, each of size $< \kappa$, so by regularity of κ , $|V_\lambda| < \kappa$. $\square$

This in turn implies we can bound the cardinality of elements of V_κ .

Proposition 0.6.

Suppose κ is strongly inaccessible and $x \subseteq V_\kappa$. Then

$$x \in V_\kappa \Leftrightarrow |x| < \kappa.$$

Proof. ($\Rightarrow$) If $x \in V_\kappa$, then $x \in V_\alpha$ for some $\alpha < \kappa$, and by transitivity $x \subseteq V_\alpha$. By the proof of Proposition 0.5, $|x| \leq |V_\alpha| < \kappa$.

($\Leftarrow$) Since $x \subseteq V_\kappa$, each $y \in x$ has $\text{rank}(y) < \kappa$. Since $|x| < \kappa$, by regularity of κ ,

$$\text{rank}(x) = \sup\{\text{rank}(y) + 1 : y \in x\} < \kappa$$

which implies $x \in V_\kappa$. $\square$

We have already seen that for limit $\alpha > \omega$, V_α is a model of all ZFC axioms except *Replacement*.

Theorem 0.2.

If κ is strongly inaccessible, then $V_\kappa \models \text{ZFC}$.

Proof. We verify that V_κ satisfies the axiom of *Replacement*. Suppose $x \in V_\kappa$ and $f : x \rightarrow V_\kappa$ is a function. Then $f[x] \subseteq V_\kappa$, and by Proposition 0.6, $|f[x]| \leq |x| < \kappa$. Applying the other direction of Proposition 0.6 to $f[x]$, we obtain $f[x] \in V_\kappa$, as desired. $\square$

Suppose an inaccessible cardinal exists, and let κ be the least inaccessible. It is not hard to verify that

$$V_\kappa \models \text{ZFC} + \text{“there does not exist an inaccessible cardinal”}.$$

(You verify that being an inaccessible cardinal is absolute for V_κ .) Therefore, the existence of an inaccessible cardinal is not provable from ZFC. This fact also follows from Gödel’s second incompleteness theorem.